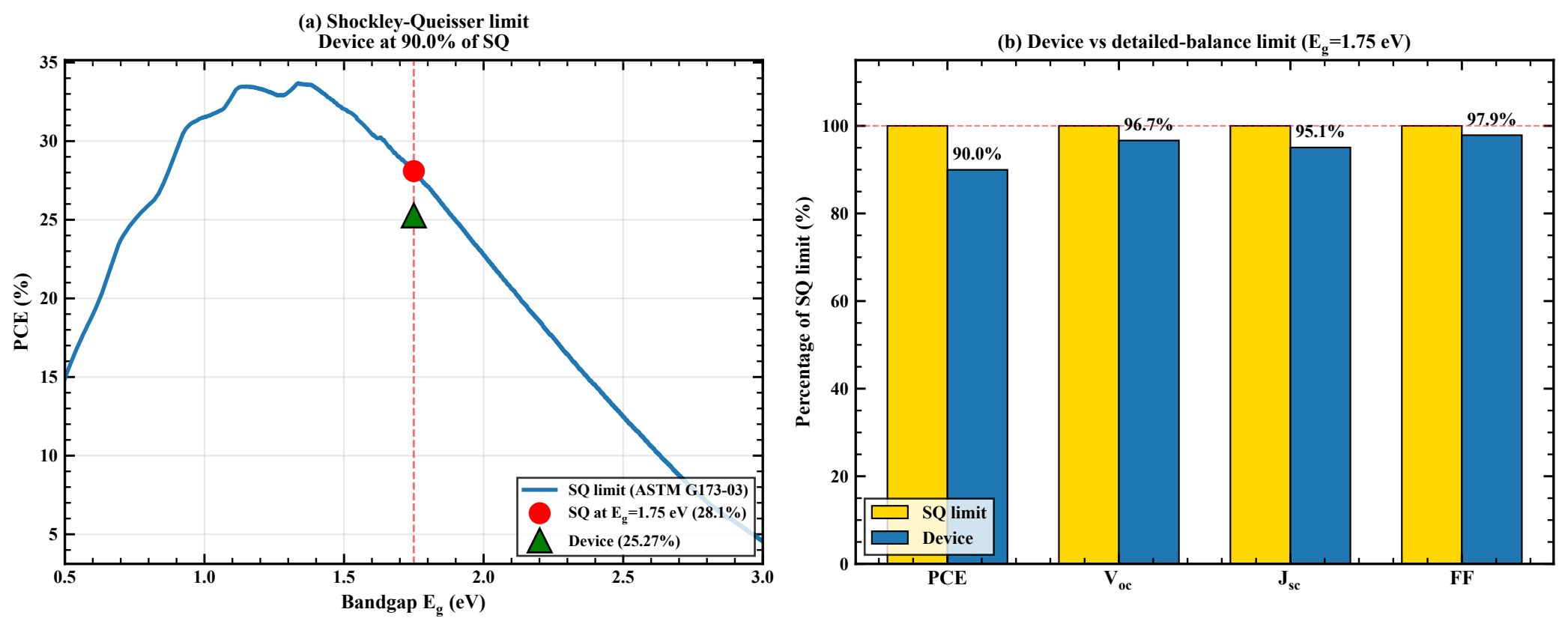
(a) Shockley-Queisser limit
Device at 90.0% of SQ
PCE (%)
Bandgap E_g (eV)
SQ limit (ASTM G173-03)
SQ at E_g=1.75 eV (28.1%)
Device (25.27%)
(b) Device vs detailed-balance limit (E_g=1.75 eV)
Percentage of SQ limit (%)
90.0%
96.7%
95.1%
97.9%
SQ limit
Device
PCE
V_{oc}
J_{sc}
FF